

PART-A

MODEL ANSWERS OF END-SEMESTER EXAMINATION (TOTAL POINTS: 20)

1. (5 points) Let Y and ϵ are the response variable and the error in a regression problem, respectively. θ_1 and θ_2 are the two parameters. Let $y_i, i = 1, 2, 3$, be the observed values of the response variable. Consider,

$$\begin{aligned} Y_1 &= \theta_1 + \theta_2 + \epsilon_1, \\ Y_2 &= \theta_1 - 2\theta_2 + \epsilon_2, \\ Y_3 &= 2\theta_1 - \theta_2 + \epsilon_3, \end{aligned}$$

where $E(\epsilon_i) = 0$, and $V(\epsilon_i) = \sigma^2, i = 1, 2, 3; \sigma > 0$. Find the least square estimates (LSE) of θ_1 and θ_2 .

Solution: We have

$$\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}_{3 \times 1} = \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}_{3 \times 2} \begin{bmatrix} \theta_1 \\ \theta_2 \end{bmatrix}_{2 \times 1} + \begin{pmatrix} \epsilon_1 \\ \epsilon_2 \\ \epsilon_3 \end{pmatrix} \Rightarrow Y = X\theta + \epsilon.$$

The least square estimator of θ is $\hat{\theta} = (X'X)^{-1}X'Y$. Now,

$$\begin{aligned} (X'X) &= \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}_{2 \times 3} \begin{bmatrix} 1 & 1 \\ 1 & -2 \\ 2 & -1 \end{bmatrix}_{3 \times 2} \\ &= \begin{bmatrix} 1+1+4=6 & 1-2-2=-3 \\ 1-2-2=-3 & 1+4+1=6 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ -3 & 6 \end{bmatrix}. \\ (X'X)^{-1} &= \frac{1}{36-9} \begin{bmatrix} 6 & 3 \\ 3 & 6 \end{bmatrix} = \frac{3}{27} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \end{aligned}$$

Thus

$$\begin{aligned} \hat{\theta} &= \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}_{2 \times 2} \begin{bmatrix} 1 & 1 & 2 \\ 1 & -2 & -1 \end{bmatrix}_{2 \times 3} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} \begin{bmatrix} y_1 + y_2 + 2y_3 \\ y_1 - 2y_2 - y_3 \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 2(y_1 + y_2 + 2y_3) + (y_1 - 2y_2 - y_3) \\ (y_1 + y_2 + 2y_3) + 2(y_1 - 2y_2 - y_3) \end{bmatrix} \\ &= \frac{1}{9} \begin{bmatrix} 3y_1 + 3y_3 \\ 3y_1 - 3y_2 \end{bmatrix} \\ &= \begin{bmatrix} \frac{1}{3}(y_1 + y_3) \\ \frac{1}{3}(y_1 - y_2) \end{bmatrix}. \end{aligned}$$

Hence, $\hat{\theta}_1 = \frac{1}{3}(y_1 + y_3)$ and $\hat{\theta}_2 = \frac{1}{3}(y_1 - y_2)$.

2. Let U_1 and U_2 be two independent uniform random variables on the interval $(0, 1)$. Suppose that $\mathbf{X} = (X_1, X_2, X_3)'$, where $X_1 = U_1$, $X_2 = U_2$, and $X_3 = U_1 + U_2$.

- (a) (3 points) Compute the correlation matrix ρ of $\mathbf{X}$.

Solution:

To compute correlation matrix ρ of X .

$$E(U_1) = E(U_2) = \frac{1}{2} \quad \text{and} \quad V(U_1) = V(U_2) = \frac{1}{12}.$$

So, $V(X_1) = V(X_2) = \frac{1}{12}$ and $V(X_3) = V(U_1) + V(U_2) = \frac{2}{12}$.

$$\begin{aligned} \text{Cov}(X_1, X_2) &= \text{Cov}(U_1, U_2) = 0 = \text{Cov}(X_2, X_1) \\ \text{Cov}(X_1, X_3) &= \text{Cov}(U_1, U_1 + U_2) \\ &= \text{Cov}(U_1, U_1) \\ &= V(U_1) = \frac{1}{12} = \text{Cov}(X_3, X_1) \\ \text{Cov}(X_2, X_3) &= \text{Cov}(U_2, U_1 + U_2) \\ &= V(U_2) = \frac{1}{12} = \text{Cov}(X_3, X_2). \end{aligned}$$

Now, $\text{Corr}(X_i, X_i) = 1$ for $i = 1, 2, 3$. Moreover,

$$\begin{aligned} \text{Corr}(X_1, X_2) &= 0, \\ \text{Corr}(X_1, X_3) &= \text{Corr}(X_3, X_1) = \frac{\text{Cov}(X_1, X_3)}{\sqrt{V(X_1)V(X_3)}} = \frac{\frac{1}{12}}{\sqrt{\frac{1}{12} \cdot \frac{2}{12}}} = \frac{1}{\sqrt{2}}, \\ \text{Corr}(X_2, X_3) &= \frac{\text{Cov}(X_2, X_3)}{\sqrt{V(X_2)V(X_3)}} = \frac{1}{\sqrt{2}}. \end{aligned}$$

Hence, the correlation matrix ρ of X is

$$\rho = \begin{bmatrix} 1 & 0 & \frac{1}{\sqrt{2}} \\ 0 & 1 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 1 \end{bmatrix}.$$

- (b) (3 points) Find the variance of first principal component based on the correlation matrix that you find in part (a).

Solution: Let Y_1 be the first principal component based on the correlation matrix ρ . Then

$$V(Y_1) = \lambda_1,$$

where λ_1 is the highest eigenvalue of the matrix ρ . The eigenvalues of ρ are solution of the following equation:

$$\begin{aligned} (1-\lambda) \begin{vmatrix} 1-\lambda & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 1-\lambda \end{vmatrix} - 0 \begin{vmatrix} 0 & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & 1-\lambda \end{vmatrix} + \frac{1}{\sqrt{2}} \begin{vmatrix} 0 & 1-\lambda \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{vmatrix} &= 0 \\ \Rightarrow (1-\lambda) \left[(1-\lambda)^2 - \frac{1}{2} \right] - 0 + \frac{1}{\sqrt{2}} \left[0 - \frac{1}{\sqrt{2}}(1-\lambda) \right] &= 0 \\ \Rightarrow (1-\lambda)^3 - \frac{1}{2}(1-\lambda) - \frac{1}{2}(1-\lambda) &= 0 \\ \Rightarrow (1-\lambda)^3 - (1-\lambda) &= 0 \\ \Rightarrow (1-\lambda)((1-\lambda)^2 - 1) &= 0 \\ \Rightarrow (1-\lambda)(\lambda^2 - 2\lambda) &= 0 \\ \Rightarrow \lambda(1-\lambda)(\lambda-2) &= 0. \end{aligned}$$

Hence, the largest eigenvalue is 2. So,

$$V(Y_1) = 2.$$

3. Let the matrix of distance be

$$\begin{array}{c} \begin{array}{cccc} & 1 & 2 & 3 & 4 \\ \begin{array}{c} 1 \\ 2 \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 \\ 1 \\ 11 \\ 5 \end{bmatrix} & \begin{bmatrix} \\ 0 \\ 2 \\ 3 \end{bmatrix} & \begin{bmatrix} \\ \\ 0 \\ 3.5 \end{bmatrix} & \begin{bmatrix} \\ \\ \\ 0 \end{bmatrix} \end{array} \end{array}$$

(a) (6 points) Cluster the four items using each of the following procedures:

- Complete linkage hierarchical procedure.
- Average linkage hierarchical procedure.

Solution: At the first stage merge objects 1 and 2 together as they are most similar.
At stage 2, we compute distances using complete linkage:

$$\begin{aligned} d_{(12),3} &= \max(d_{13}, d_{23}) = \max(11, 2) = 11, \\ d_{(12),4} &= \max(d_{14}, d_{24}) = \max(5, 3) = 5. \end{aligned}$$

So, we obtain the distance matrix,

$$\begin{array}{c} \begin{array}{ccc} & (12) & 3 & 4 \\ \begin{array}{c} (12) \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 \\ 11 \\ 5 \end{bmatrix} & \begin{bmatrix} \\ 0 \\ 3.5 \end{bmatrix} & \begin{bmatrix} \\ \\ 0 \end{bmatrix} \end{array} \end{array}$$

At stage 3, merge objects 3 & 4, as they are most similar.
Now, we compute,

$$\begin{aligned} d_{(12)(34)} &= \max\{d_{(12)3}, d_{(12)4}\} \\ &= \max\{11, 5\} \\ &= 11 \end{aligned}$$

So, we obtain the new distance matrix,

$$\begin{array}{c} \begin{array}{cc} & (12) & (34) \\ \begin{array}{c} (12) \\ (34) \end{array} & \begin{bmatrix} 0 \\ 11 \end{bmatrix} & \begin{bmatrix} \\ 0 \end{bmatrix} \end{array} \end{array}$$

Now, at the final stage, merge clusters (12) & (34) together to form a single cluster.

jaa

At the first stage merge objects 1 and 2 together as they are most similar.
At stage 2, we compute distances using complete linkage:

$$\begin{aligned} d_{(12),3} &= \frac{d_{13} + d_{23}}{2} = \frac{11 + 2}{2} = 6.5, \\ d_{(12),4} &= \frac{d_{14} + d_{24}}{2} = \frac{5 + 3}{2} = 4. \end{aligned}$$

So, we obtain the distance matrix,

$$\begin{array}{c} \begin{array}{ccc} & (12) & 3 & 4 \\ \begin{array}{c} (12) \\ 3 \\ 4 \end{array} & \begin{bmatrix} 0 \\ 6.5 \\ 4 \end{bmatrix} & \begin{bmatrix} \\ 0 \\ 3.5 \end{bmatrix} & \begin{bmatrix} \\ \\ 0 \end{bmatrix} \end{array} \end{array}$$

At stage 3, merge objects 3 & 4, as they are most similar.

Now, we compute,

$$d_{(12)(34)} = \frac{d_{(12)3} + d_{(12)4}}{2} = \frac{6.5 + 4}{2} = 5.25.$$

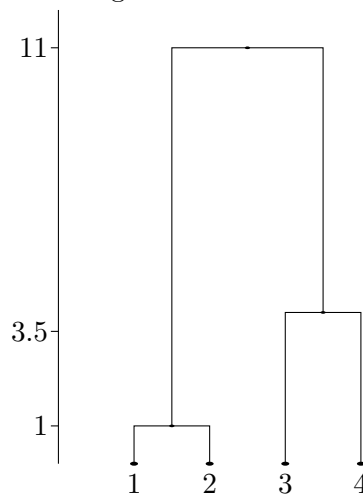
So, we obtain the new distance matrix,

$$\begin{array}{cc} & \begin{matrix} (12) & (34) \end{matrix} \\ \begin{matrix} (12) \\ (34) \end{matrix} & \begin{bmatrix} 0 & \\ 5.25 & 0 \end{bmatrix} \end{array}$$

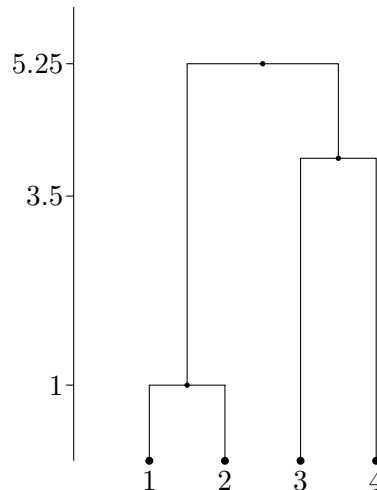
Now, at the final stage, merge clusters (12) & (34) together to form a single cluster.

- (b) (3 points) Draw the dendrograms and compare the results obtained using previous two procedures.

Solution: Dendrogram for complete linkage:



Dendrogram for average linkage:



Comparing the two dendrograms, we observe that the single linkage and complete linkage do not differ in allocating objects in this case.